

# Discussion 11 Worksheet

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## Formula Bank

- **Probability:** For an event  $A$  in a sample space  $S$ ,  $P(A) = \frac{|A|}{|S|}$
- **Conditional Probability:** The probability of  $A$  given  $B$ ,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- **Law of Total Probability:** If  $\{B_1, B_2, \dots, B_n\}$  is a partition of the sample space,

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i)$$

- **Bayes' Rule:**

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{P(A)} = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^n P(A | B_j)P(B_j)}$$

- **Expectation:** For a discrete random variable  $X$ ,  $\mathbb{E}[X] = \sum_x x P(X = x)$

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## Probability

1. A committee of 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?

2. A 5-card poker hand is said to be a full house if it consists of 3 cards of the same denomination and 2 other cards of the same denomination (of course, different from the first denomination). Thus, one kind of full house is three of a kind plus a pair. What is the probability that one is dealt a full house?
3. An elementary school is offering 3 language classes: one in Spanish, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 28 students in the Spanish class, 26 in the French class, and 16 in the German class. There are 12 students that are in both Spanish and French, 4 that are in both Spanish and German, and 6 that are in both French and German. In addition, there are 2 students taking all 3 classes.
- (a) If a student is chosen randomly, what is the probability that the student is not in any of the language classes?
- (b) If a student is chosen randomly, what is the probability that the student is taking exactly one language class?

## Conditional probability:

1. In Section 1, problem 3, what is the probability that a student selected at random is taking French, given that the student is also taking Spanish?
2. A family has two children. It is known that at least one of them is a boy. What is the probability that the other child is a girl?
3. A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a “false positive” result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

## Expectation:

1. A restaurant runs a promotion: every customer draws one card from a shuffled deck of 10 cards.
  - 2 cards give a 50% discount,
  - 3 cards give a 20% discount,
  - 5 cards give no discount.

Emily's meal costs \$48 before any discount. Compute the expected amount she pays.

2. A child asks his parents for some money. The parents make the following offers.
  - Father's offer: The child rolls a fair die. He will pay the child \$4 times the number rolled.
  - Mother's offer: The child flips a fair coin. If it lands heads, she pays the child \$22. If it lands tails, she pays nothing.

Whose offer should the child consider?